1.1 Structure of DNA

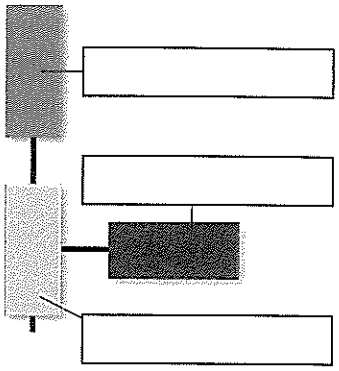
Science understanding

Visual/Spatial Logical/Mathematical

DNA building blocks

The building blocks of the DNA molecule are nucleotides. Each nucleotide has three parts:

- phosphate group
- sugar
- nitrogen-rich base.



1 **Use** these three terms to label the nucleotide shown right.

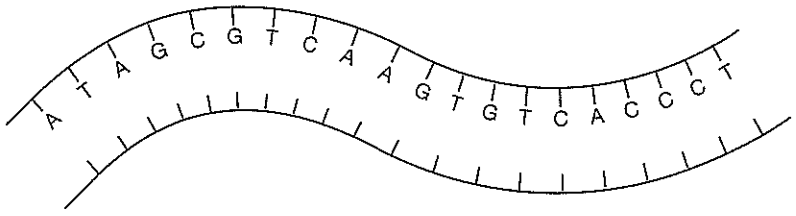
DNA structure

The nitrogen-rich bases can be one of four types:

- adenine (A)
- thymine (T)
- cytosine (C)
- guanine (G).

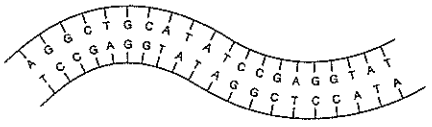
In a double-stranded DNA molecule, A and T always form a pair and C and G always form a pair.

2 The sequence of bases below represents a section of a single strand of DNA. **Propose** the base sequence in the complementary strand of DNA.

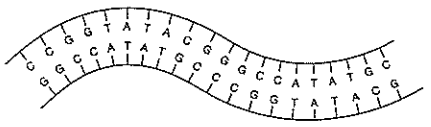


3 (a) **Identify** which of the following figures represents a possible base sequence in a molecule of DNA. _____

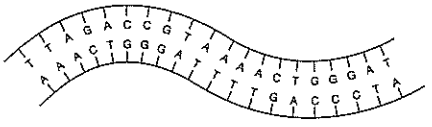
(i)



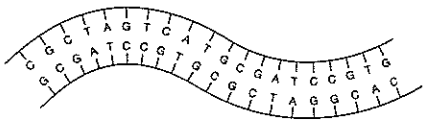
(ii)



(iii)



(iv)



(b) **Justify** your decision for each molecule.

(i) _____

(ii) _____

(iii) _____

(iv) _____

1.2 Discovery of DNA

Science as a human endeavour

Verbal/Linguistic

Refer to the Science as a Human Endeavour on pages 5–6 of your student book to answer the following questions.

1 (a) **State** when the chemical now known as DNA was first identified.

_____

(b) **Name** the scientist who first identified it.

_____

2 **Describe** the contribution that Phoebus Levene made to what was known about DNA.

_____

_____

3 **Explain** why Levene did not think that DNA was the chemical that carried the genetic code.

_____

_____

4 **Identify** the scientists who showed Levene's reasoning to be incorrect.

_____

5 **Identify** the invention that led to the discovery of the structure of DNA.

_____

6 Erwin Chagraff could not explain why the amount of guanine plus adenine was always equal to the amount of thymine plus cytosine in DNA. In your own words, **explain** why this relationship is always found.

_____

_____

_____

7 **Construct** a timeline from 1869 to 1965 showing the significant events leading to the scientific understanding of the structure of DNA.

